STRUCTURAL ANALYSIS 1 (INS2512) - LESSON 5 -

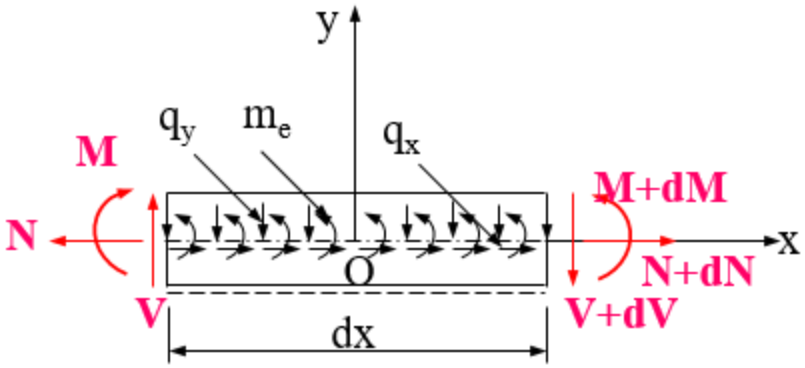
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Analysis of Coplanar Systems Under Dead Loads

The Relationships between External Loads and Internal Forces

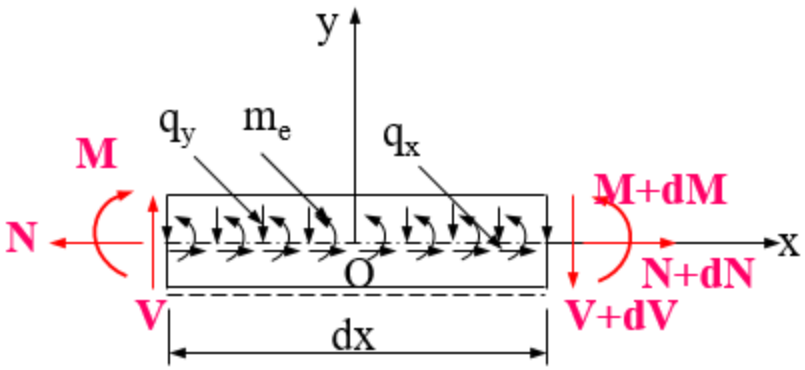
- ***Distributed Loads***

Consider a differential element with a length of dx subjected to the distributed load «qx» parallel to its axis, the distributed load «qy» perpendicular to its axis and the distributed moment «me».

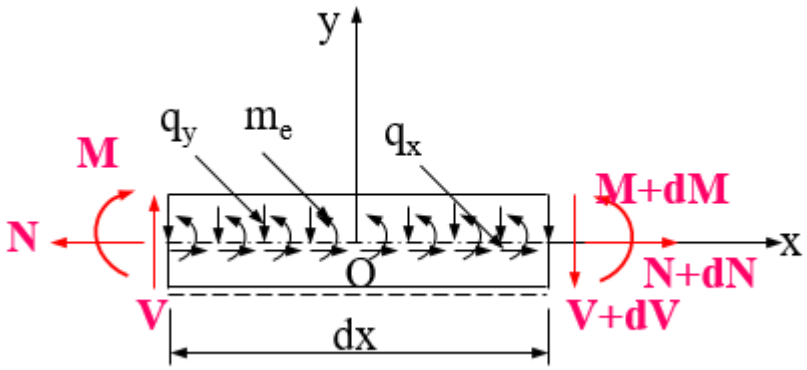


Positive signs:

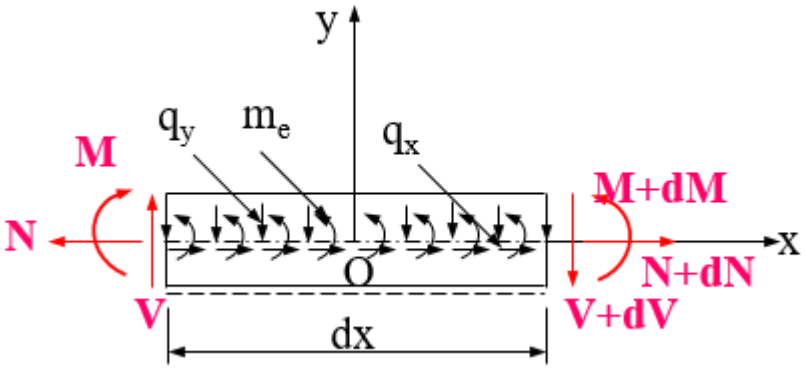
- qx: acts from left to right
- qy: acts downwards
- me: acts in the counter-clock wise



The bottom side of the element is marked for identifying the sign convention of internal forces.



Hence, the internal forces at the left end of the element is N, V and M, and the internal forces at the right end of the member is N+dN, V+dV and M+dM. Let's write the equations of equilibrium:

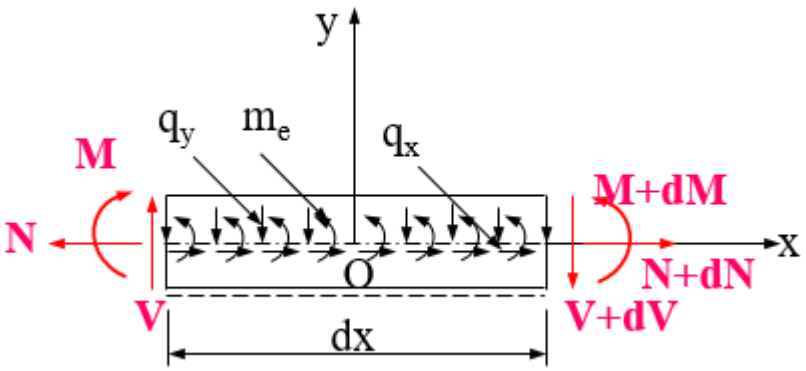


Let's write the equations of equilibrium:

$$\sum P_x = 0$$

$$-N + N + dN + q_x dx = 0 \quad \Rightarrow \quad dN + q_x dx = 0$$

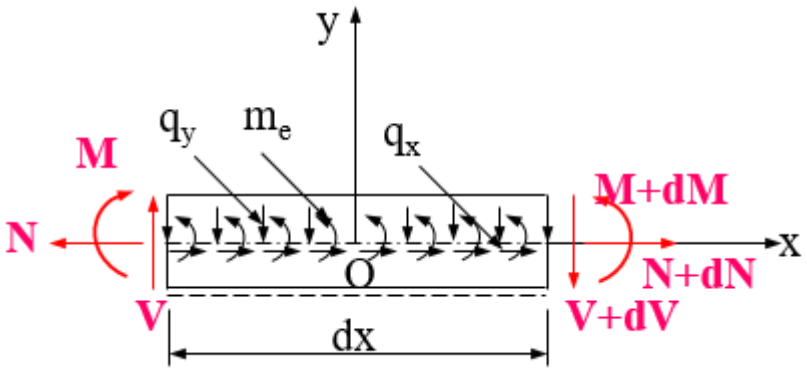
$$\frac{dN}{dx} + q_x = 0 \quad \text{or} \quad \frac{dN}{dx} = -q_x$$



$$\sum P_y = 0$$

$$-V + V + dV + q_y dx = 0 \quad \Rightarrow \quad dV + q_y dx = 0$$

$$\frac{dV}{dx} + q_y = 0 \qquad \text{or} \qquad \frac{dV}{dx} = -q_y$$

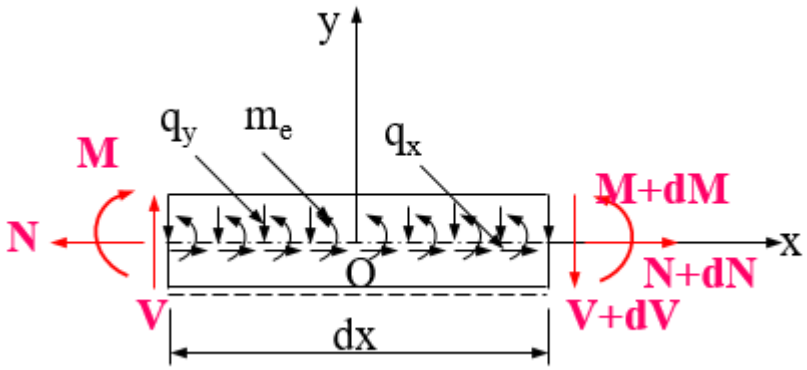


$$\sum M_0 = 0$$

$$M - (M + dM) + V\frac{dx}{2} + (V + dV)\frac{dx}{2} - m_e dx = 0$$

$$-dM + Vdx - m_e dx = 0 \qquad \left(\frac{dVdx}{2} \approx 0\right)$$

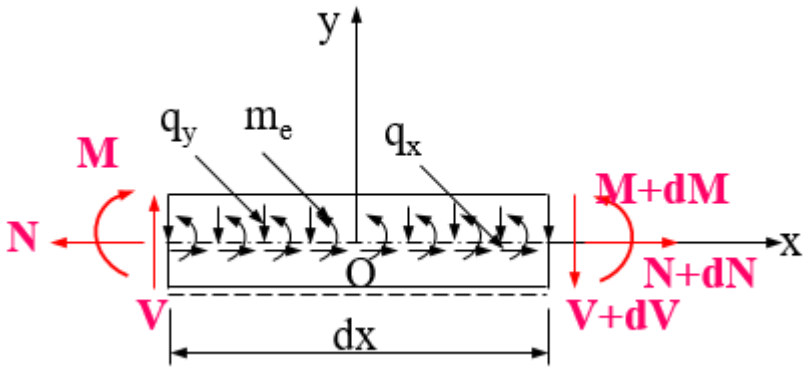
$$\frac{dM}{dx} - V + m_e = 0 \quad \text{or} \quad \frac{dM}{dx} = V - m_e$$



$$\frac{dM}{dx} - V + m_e = 0$$ or $$\frac{dM}{dx} = V - m_e$$

A distributed moment is not a common type of load in practice, so it can be taken as 0. Then,

$$\frac{dM}{dx} - V = 0$$ or $$\frac{dM}{dx} = V$$



$$\frac{dM}{dx} - V = 0$$ or $$\frac{dM}{dx} = V$$

If this expression is differentiated with respect to x,

$$\frac{d^2M}{dx^2} = \frac{dV}{dx} = -q_y$$

the relationship between the external load and the internal moment is established.

Examining the relationships between the external loads and internal forces, the following remarks can be stated:

- The load applied paralled to the axis of the element (qx) does not cause any bending moment or shear force. It causes only axial force. The slope (or tangent) of the axial force diagram of the element at any section is equal to the magnitude of the external load parallel to the axis of the element (qx) at the same section.

$$\frac{dN}{dx} = -q_x$$

- The load applied perpendicular to the axis of the element (qy) causes both bending moment and shear force, however it does not cause any axial force. The opposite of the slope (or tangent) of the shear force diagram of the element at any section is equal to the magnitude of the external load perpendicular to the axis of the element (qy) at the same section.

$$\frac{dV}{dx} = -q_y$$

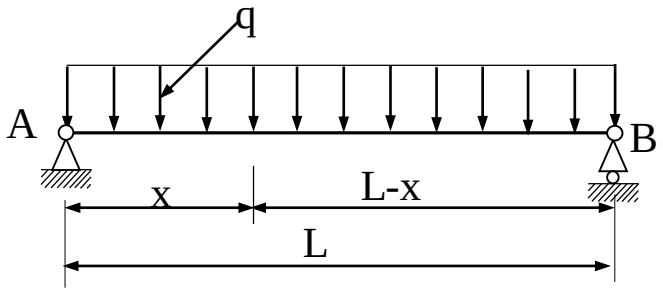
Slope of Shear diagram	=	Magnitude of distributed load

- The slope (or tangent) of the bending moment diagram of the element at any section is equal to the magnitude of the shear force at the same section.

$$\frac{dM}{dx} = V$$

Slope of Moment diagram = Shear

Example:



Derive the expressions for the shear force and the bending moment.

Solution:

$$\frac{d^2M}{dx^2} = -q$$

Integrate with respect to x:

$$\frac{dM}{dx} = -\int q \cdot dx = -q \cdot x + C_1 \qquad \frac{dM}{dx} = V(x) \longrightarrow V_x = -q \cdot x + C_1$$

Integrate once more with respect to x:

$$M_x = \int (-q \cdot x + C_1)dx = -qx^2/2 + C_1 x + C_2$$

Boundary conditions to estimate C_1 and C_2

$$x = 0 \longrightarrow M_{AB} = 0 \qquad x = L \longrightarrow M_{BA} = 0$$

$$M_{AB} = -q.(0)^2/2 + C_1(0) + C_2 = 0 \longrightarrow C_2 = 0$$

$$M_{BA} = -q.L^2/2 + C_1 L + C_2 = 0 \longrightarrow C_1 = qL/2$$

Expression for shear force:

$$V_x = q\frac{L}{2} - qx = q(\frac{L}{2} - x)$$

$$x = 0 \longrightarrow V_{AB} = qL/2$$

$$x = L \longrightarrow V_{BA} = -qL/2$$

Expression for bending moment:

$$M_x = q\frac{L}{2}x - q\frac{x^2}{2} = \frac{q}{2}x(L - x)$$

$$x=0 \longrightarrow M_{AB}=0$$

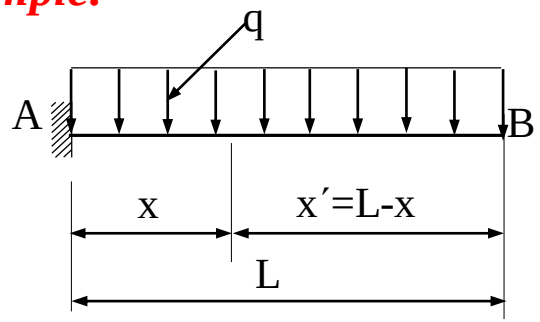
$$x=L \longrightarrow M_{BA}=0$$

The maximum moment is observed at the location where V = 0. Thus;

$$V_x = q(\frac{L}{2} - x) = 0 \longrightarrow x_0 = L/2$$

$$M_{enb.} = q\frac{L}{2}\cdot\frac{L}{2} - q\frac{(L/2)^2}{2} \longrightarrow M_{enb.} = q\frac{L^2}{8}$$

Example:



Derive the expressions for the shear force and the bending moment.

Solution:

Since the load is constant (uniform distributed load), V(x) and M(x) have the same form as those do in the previous example:

$$V_x = -q \cdot x + C_1 \qquad M_x = -qx^2/2 + C_1 x + C_2$$

Boundary conditions to estimate C_1 and C_2

Since B is a free end and there is no point load at this end:

$$x = L \longrightarrow V_{BA} = 0 \qquad x = L \longrightarrow M_{BA} = 0$$

Then,

$$V_{BA} = -q.L + C_1 = 0 \longrightarrow C_1 = qL$$

$$M_{BA} = -q.L^2/2 + C_1 L + C_2 = 0 \longrightarrow C_2 = -qL^2/2$$

Expression for shear force:

$$V_x = qL - qx = q(L - x)$$

$$V_x = qx'$$

$$x = 0 \longrightarrow V_{AB} = qL$$

$$x = L \longrightarrow V_{BA} = 0$$

Expression for bending moment:

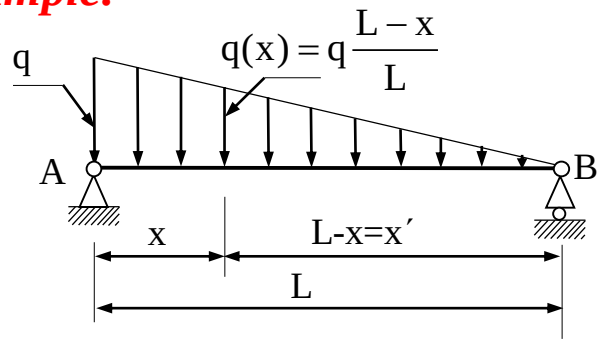
$$M_x = -q\frac{x^2}{2} + qLx - qL^2/2 = -\frac{q}{2}(L - x)^2$$

$$M_x = -qx'^2/2$$

$$x = 0 \longrightarrow M_{AB} = -qL^2/2$$

$$x = L \longrightarrow M_{BA} = 0$$

Example:



Derive the expressions for the shear force and the bending moment.

Solution:

$$\frac{dV}{dx} = -(qx) = -q\frac{L-x}{L}$$

Integrate with respect to x:

$$V_x = -q\int \frac{L-x}{L}dx \longrightarrow V_x = -q\frac{(Lx - x^2/2)}{L} + C_1$$

$$\frac{dM}{dx} = V \longrightarrow \frac{dM}{dx} = V_x = -q\frac{(Lx - x^2/2)}{L} + C_1$$

Integrate once more with respect to x:

$$M_x = \int[-q\frac{(Lx - x^2/2)}{L}dx + C_1 dx] \longrightarrow M_x = -qx^2\frac{(3L-x)}{6L} + C_1 x + C_2$$

Boundary conditions to estimate C_1 and C_2

$$x = 0 \longrightarrow M_{AB} = 0$$

$$x = L \longrightarrow M_{BA} = 0$$

$$M_{AB} = -q \cdot (0)^2 \frac{3L - 0)}{6L} + C_1 \cdot 0 + C_2 = 0$$

$$M_{BA} = -qL^2 \frac{3L - L}{6L} + C_1 L + C_2 \longrightarrow C_1 = qL/3 \qquad C_2 = 0$$

Expression for shear force:

$$V_x = \frac{qL}{6}[2 - 6\frac{x}{L} + 3(\frac{x}{L})^2]$$

$$x = 0 \longrightarrow V_{AB} = qL/3$$

$$x = L \longrightarrow V_{BA} = -qL/6$$

Expression for bending moment:

$$M_x = \frac{qL}{6}[2\frac{x}{L} - 3(\frac{x}{L})^2 + (\frac{x}{L})^3]$$

$$x = 0 \longrightarrow M_{AB} = 0$$

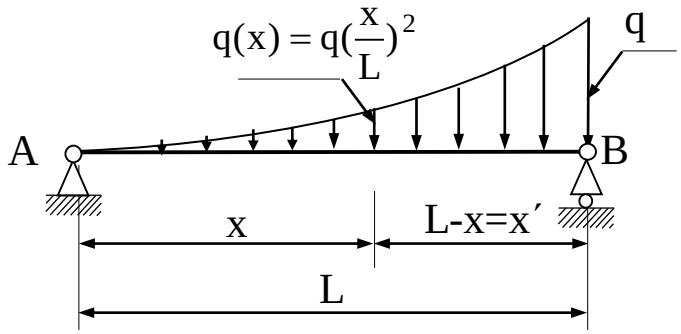
$$x = L \longrightarrow M_{BA} = 0$$

The maximum moment is observed at the location where V = 0. Thus;

$$V_x = \frac{qL}{6}[2 - 6\frac{x}{L} + 3(\frac{x}{L})^2] = 0 \longrightarrow x_0 = (1 - \frac{\sqrt{3}}{3})L \cong 0{,}42265L$$

$$M_{max} = \frac{\sqrt{3}}{27} qL^2 \cong 0{,}06415qL^2$$

Example:



Derive the expressions for the shear force and the bending moment.

Solution:

$$q_y = q(x) = q\left(\frac{x}{L}\right)^2 \qquad \frac{dV}{dx} = -q_y \qquad \frac{dV}{dx} = -q\left(\frac{x}{L}\right)^2$$

$$\frac{dM}{dx} = V \qquad \frac{dM}{dx} = V_x = -q\frac{x^3}{3L^2} + C_1 \qquad M_x = \int\left(-q\frac{x^3}{3L^2}\cdot dx + C_1 \cdot dx\right)$$

$$M_x = -q\frac{x^4}{12L^2} + C_1 x + C_2$$

Boundary conditions to estimate C_1 and C_2

$$x = 0 \longrightarrow M_{AB} = 0 \qquad x = L \longrightarrow M_{BA} = 0$$

$$M_{AB} = -q\frac{(0)^4}{12L^2} + C_1.(0) + C_2 = 0$$

$$M_{BA} = -q\frac{L^4}{12L^2} + C_1 L + C_2 = 0 \qquad C_1 = qL/12 \qquad C_2 = 0$$

Expression for shear force:

$$V_x = \frac{qL}{12}[1 - 4(\frac{x}{L})^3]$$

$$x = 0 \longrightarrow V_{AB} = qL/12$$

$$x = L \longrightarrow V_{BA} = -qL/4$$

$$V_x = \frac{qL}{12}[1 - 4(\frac{x}{L})^3] = 0$$

Expression for bending moment:

$$M_x = \frac{qL^2}{12}[\frac{x}{L} - (\frac{x}{L})^4]$$

$$x = 0 \longrightarrow M_{AB} = 0$$

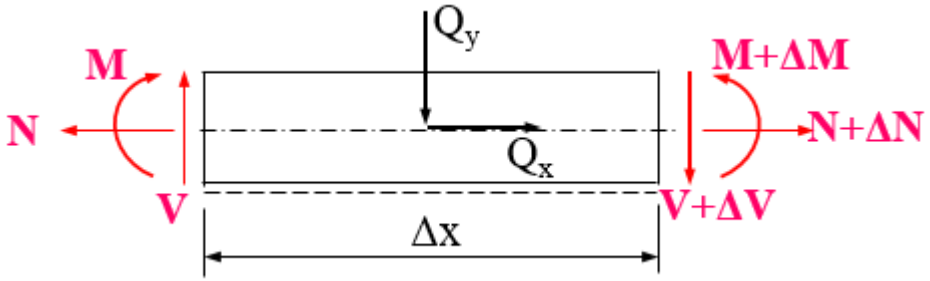
$$x = L \longrightarrow M_{BA} = 0$$

$$x_0 = \sqrt[3]{1/4} \times L \cong 0{,}62996L$$

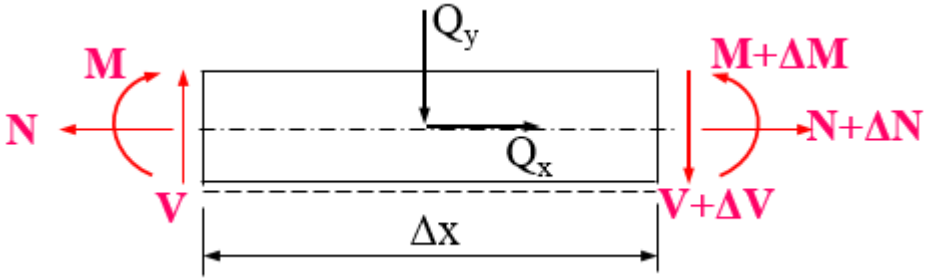
$$M_{max} = \frac{qL}{16\sqrt[3]{4}} \cong 0{,}0393725qL^2$$

- ***Point Loads***

Consider a differential element with a length of Δx subjected to a point load Qx parallel to the axis of the element and a point load Qy perpendicular to the axis of the element.



The bottom side of the element is marked for identifying the sign convention of internal loads.



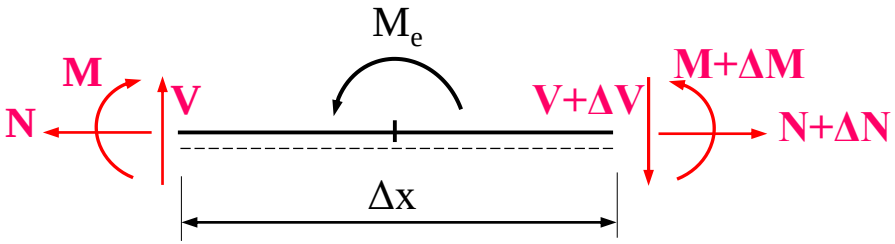
The internal forces at the left end of the element are N, V and M. Then the internal forces at the right end of the element are N+ΔN, V+ ΔV and M+ ΔM. Let's write the equations of equilibrium

$$\sum \boldsymbol{P_x} = \boldsymbol{0} \rightarrow -\boldsymbol{N} + (\boldsymbol{N} + \Delta\boldsymbol{N}) + \boldsymbol{Q_x} = \boldsymbol{0} \rightarrow \Delta\boldsymbol{N} = -\boldsymbol{Q_x}$$

$$\sum \boldsymbol{P_y} = \boldsymbol{0} \rightarrow -\boldsymbol{V} + (\boldsymbol{V} + \Delta\boldsymbol{V}) + \boldsymbol{Q_y} = \boldsymbol{0} \rightarrow \Delta\boldsymbol{V} = -\boldsymbol{Q_y}$$

$$\Delta\boldsymbol{x} \rightarrow \boldsymbol{0} \quad \Longrightarrow \quad \Delta\boldsymbol{M} = \boldsymbol{0}$$

Consider a differential element subjected to a moment Me



$$\sum \boldsymbol{P_x = 0 \rightarrow \Delta N = 0}$$

$$\sum \boldsymbol{P_y = 0 \rightarrow \Delta V = 0}$$

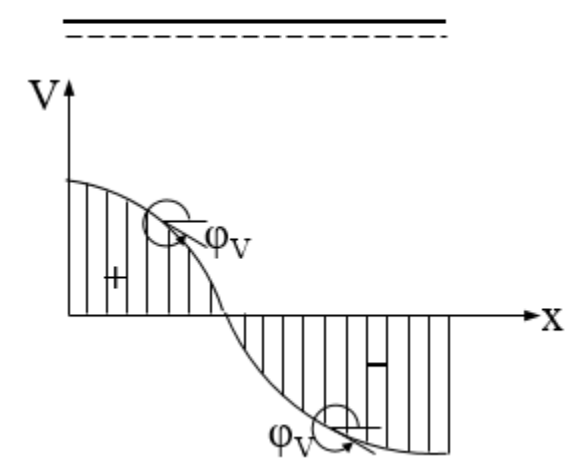
$$\sum \boldsymbol{M = 0 \rightarrow \Delta M = -M_e}$$

Examining the relationships between the external and internal loads, the following remarks can be stated:

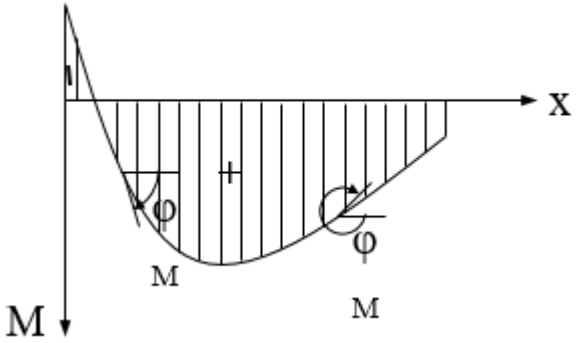
- The load applied paralled to the axis of the element (Qx) does not cause any change in the bending moment or shear force. It causes sudden change in axial force and the magnitude of this change is equal to the opposite of the load applied parallel to the axis of the element
- The load perpendicular to the axis of the element (Qy) does not cause any change in the axial force and the bending moment. It causes a sudden change in the shear force and the magnitude of this change is equal to the opposite of the load applied perpendicular to the axis of the element (Qy)

- The bending moment (Me) acting at a section of the element does not cause any change in the axial force and the shear force at the same section. It causes a sudden change in the bending moment and the magnitude of this change is equal to the opposite of the bending moment applied at the section.

Internal Force Diagrams



Shear force diagram
(Same with axial force)



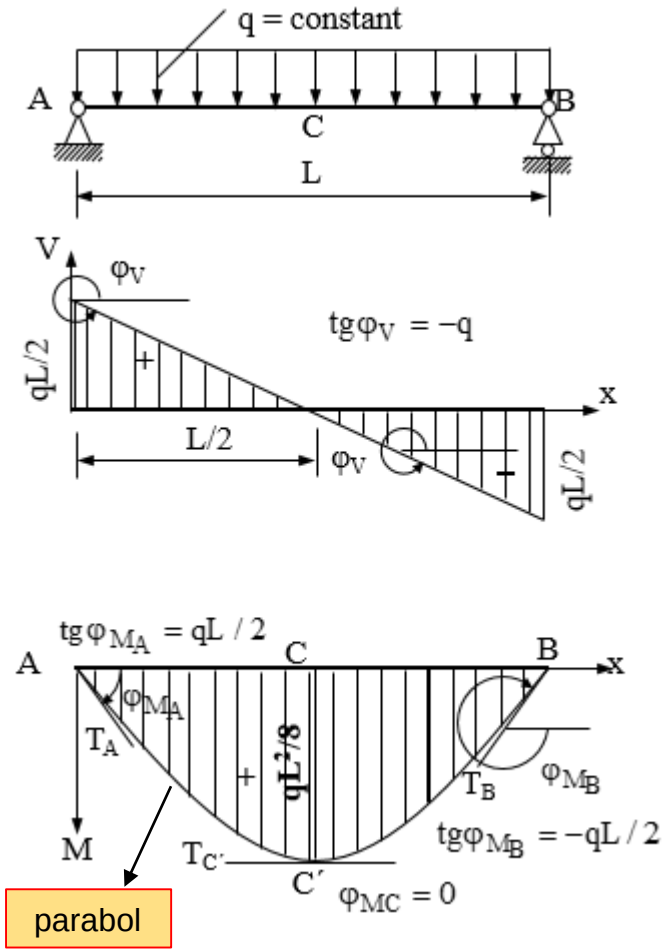
Moment diagram

In general, the positive values of the bending moment diagram are shown on the side of dashed line, while the positive values of the shear force and axial force diagrams are shown on the opposite side of the dashed line.

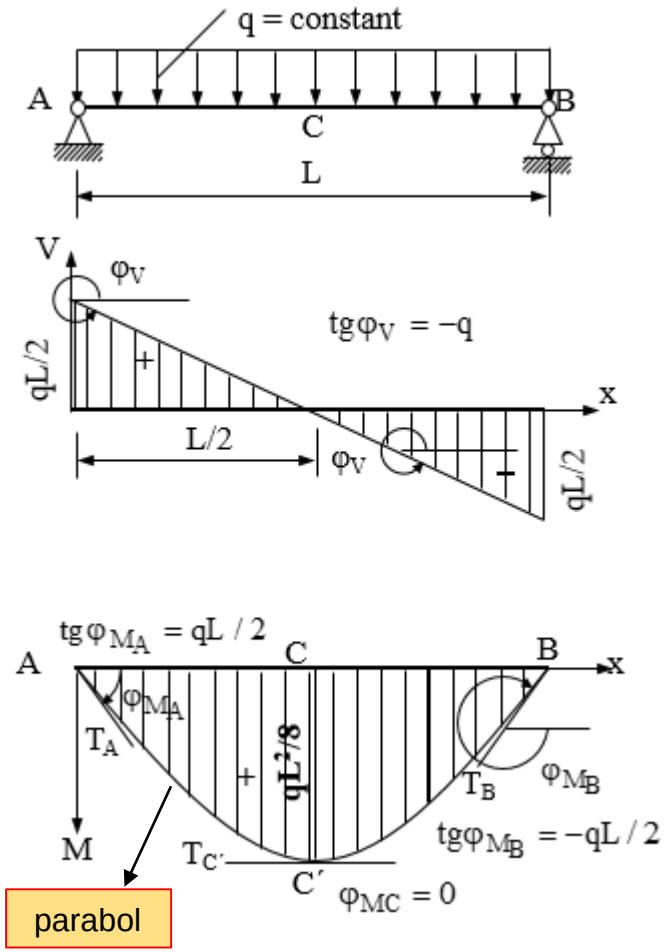
$$\frac{dN}{dx} = -q_x \longrightarrow tan\varphi_N = -q_x$$

$$\frac{dV}{dx} = -q_y \longrightarrow tan\varphi_V = -q_y$$

$$\frac{dM}{dx} = V \longrightarrow tan\varphi_M = V$$

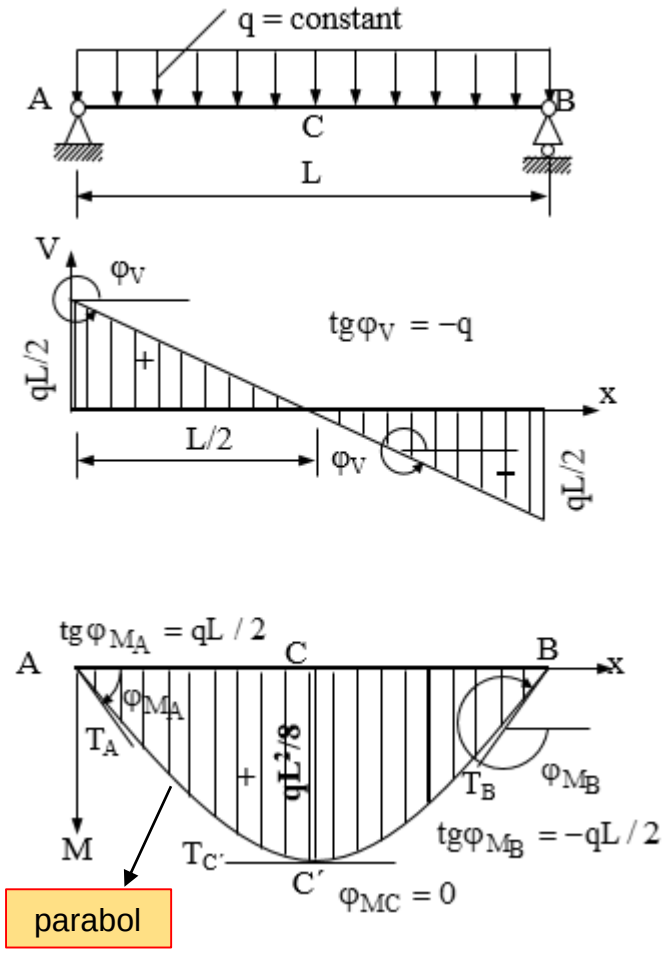


- The change in the shear force is directly related to the external load acting perpendicular to the axis of the element.
- The change in the bending moment is primarily related to the shear force.



- Since the external load is constant, the change in the shear force is linear.

Since the support reactions are known to be equal to qL/2 and -qL/2 at the supports A and B, respectively, a line is drawn between these two points to obtain the shear force diagram.



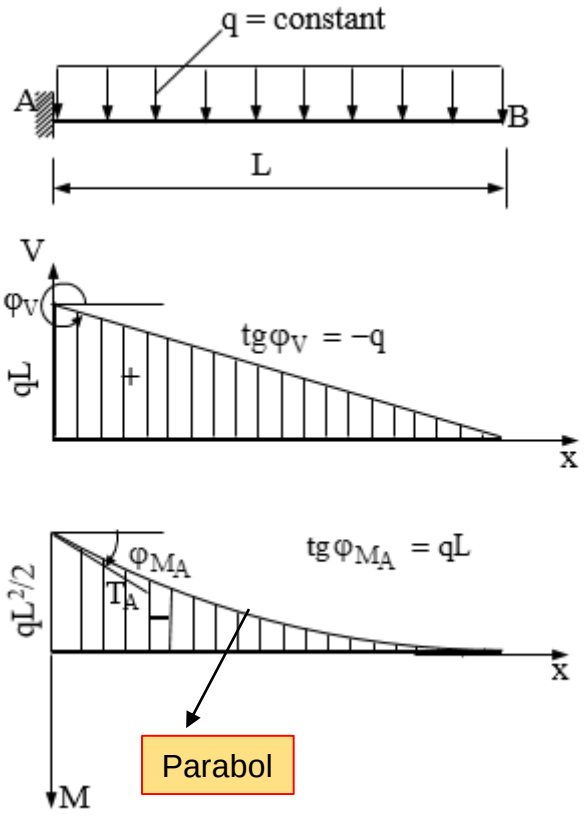
- Since the shear force is linear, the bending moment diagram is parabol.

In order to draw a parabol, three points are needed.

- At point A and B: moment is zero
- At point C: moment is maximum

Slope:

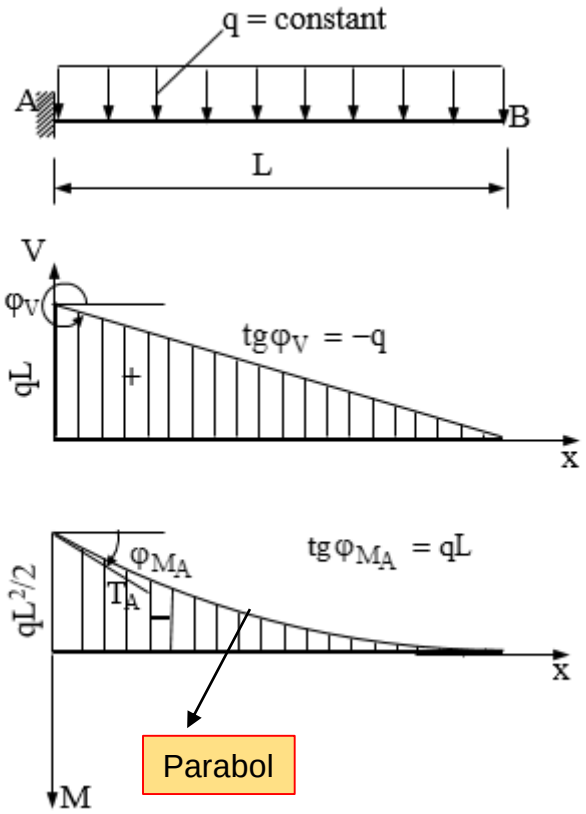
- At point A and B: slope is equal to qL/2
- At point C: slope is equal to 0
- The slope is positive between A and C since the shear force is positive
- The slope is negative between C and B since the shear force is negative



- Since the external load is constant, the change in the shear force is linear.

Since the support reaction at point A is equal to qL and the shear force at point be is equal to 0, a line is drawn between these two points to obtain shear force diagram.

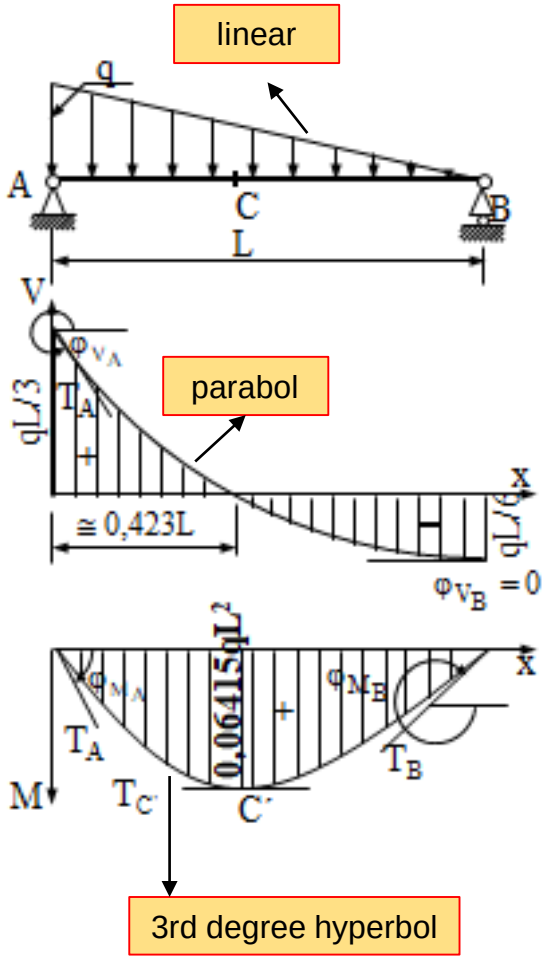
The slope of this line is equal to –q.



- Since the shear load is linear, the bending moment diagram is parabol.
- At point A: the value of the parabol is equal to $qL^2/2$
- At point B: the value of the parabol is equal to 0

Slope:

- At point A: the slope is equal to qL
- At point B: the slope is equal to 0



- Since the external load is linear, the shear force is a parabol.

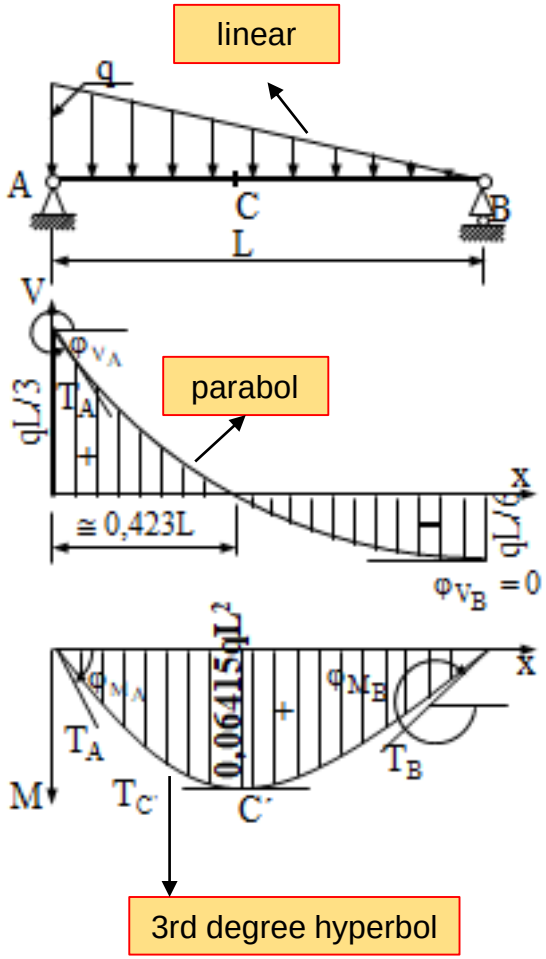
Shear forces at three sections are known:

- At point A: the value is qL/3
- At point B: the value is qL/6
- At point C: the value is 0

Slope: (tgφ)

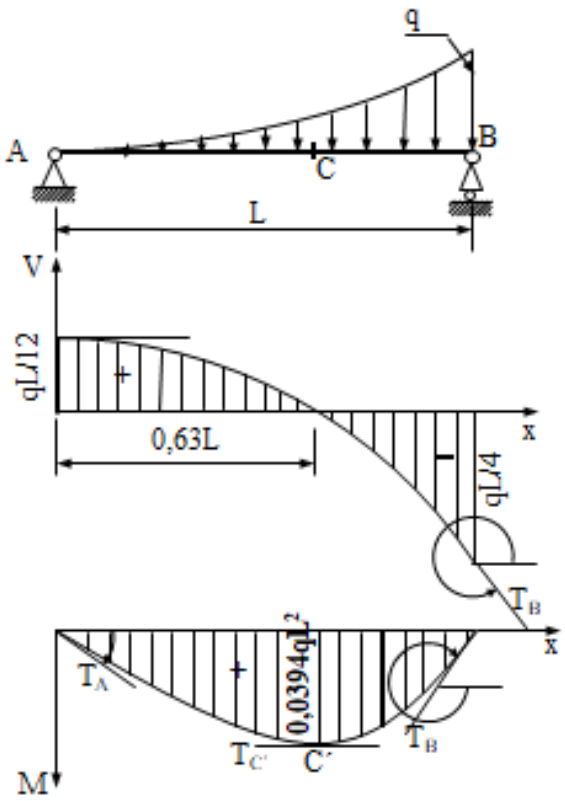
- At point A: the slope is –q
- At point B: the slope is 0

A parabolic curve, which is equal to 0 at C, is drawn from A to B.

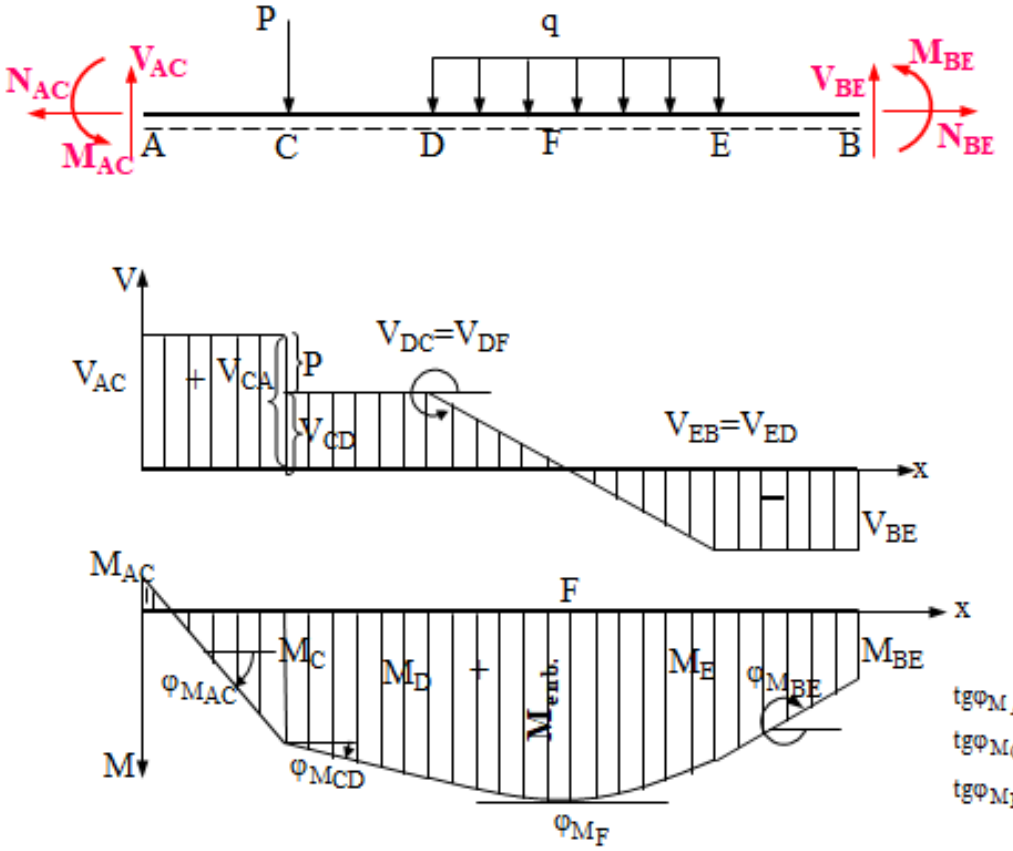


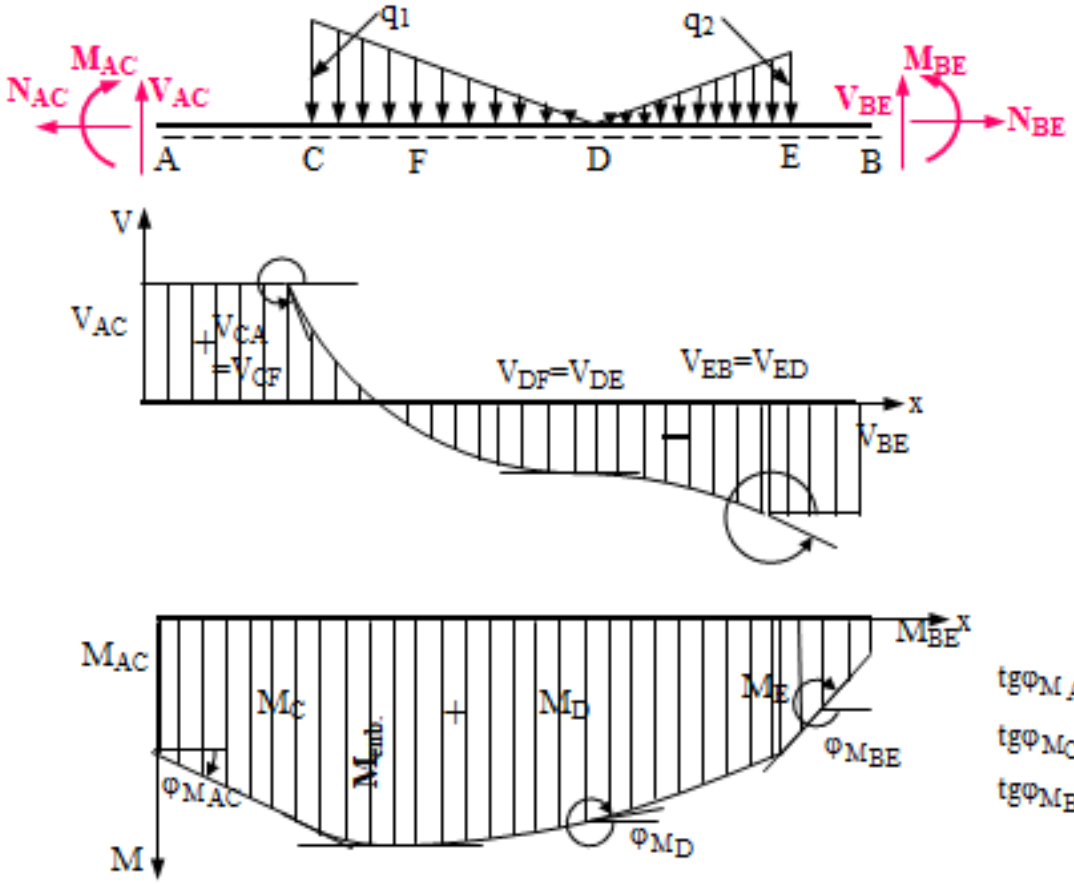
- Since the shear force is parabolic, the bending moment is a 3rd degree hyperbol.

Bending moment diagram is drawn using same way with the shear force.



- Since the external load is parabolic, the shear force diagram is 3rd degree hyperbolic.
- Since the shear force diagram is 3rd degree hyperbolic, the bending moment diagram is 4th degree hyperbolic.

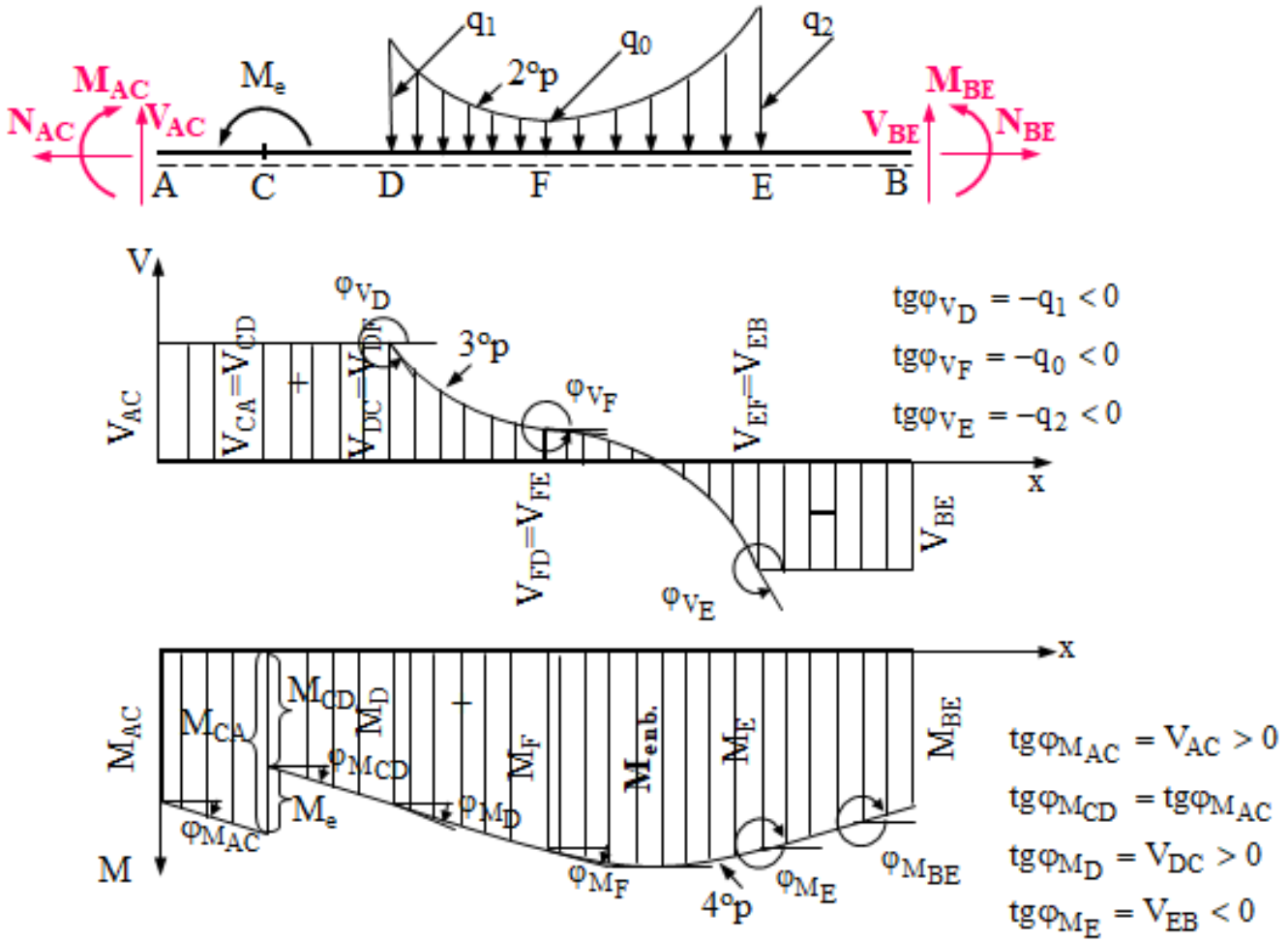

P
q
N_{AC}
V_{AC}
M_{AC}
A
C
D
F
E
B
V_{BE}
M_{BE}
N_{BE}
V
V_{AC}
+
V_{CA}
P
$V_{DC}=V_{DF}$
V_{CD}
$V_{EB}=V_{ED}$
x
V_{BE}
M_{AC}
F
x
M_C
M_D
+
M_{max}
M_E
M_{BE}
$\varphi_{M_{AC}}$
$\varphi_{M_{CD}}$
$\varphi_{M_{BE}}$
M
φ_{M_F}
$tg\varphi_{M_{AC}} = V_{AC} > 0$
$tg\varphi_{M_{CD}} = V_{CD} = V_{DC} > 0$
$tg\varphi_{M_{BE}} = V_{EB} = V_{BE} < 0$

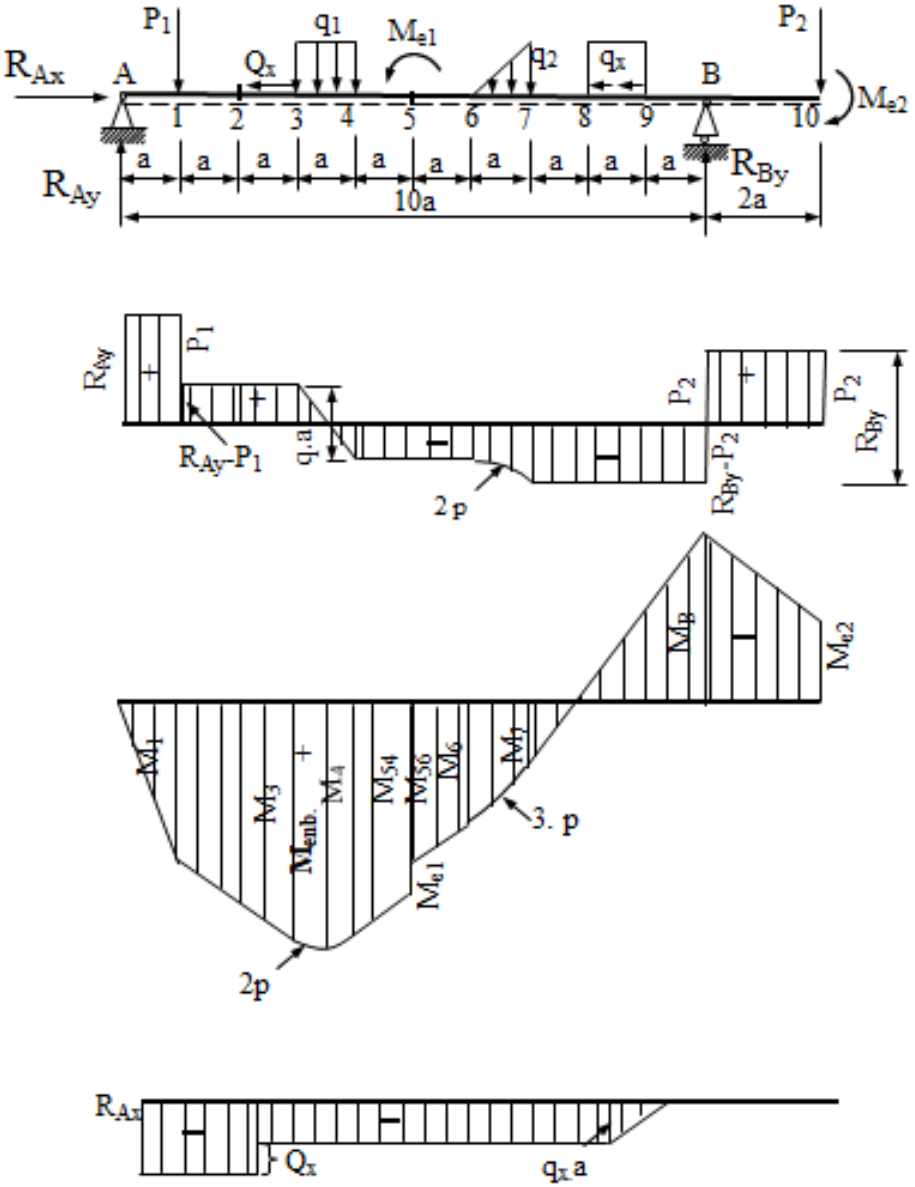


$tg\varphi_{M\,AC} = V_{AC} = V_{CA} > 0$

$tg\varphi_{M\,CD} = V_{DF} = V_{DE} < 0$

$tg\varphi_{M\,BE} = V_{EB} = V_{ED} < 0$


q1
q0
q2
2°p
MAC
Me
MBE
NAC
VAC
VBE
NBE
A
C
D
F
E
B
V
φVD
3°p
φVF
VAC
VCA=VCD
VDC=VDF
VEF=VEB
VFD=VFE
VBE
φVE
x
tgφVD = −q1 < 0
tgφVF = −q0 < 0
tgφVE = −q2 < 0
MAC
MCA
MCD
MD
φMCD
Me
φMAC
φMD
MF
Menb.
ME
MBE
φMF
4°p
φME
φMBE
M
tgφMAC = VAC > 0
tgφMCD = tgφMAC
tgφMD = VDC > 0
tgφME = VEB < 0


P1
q1
Me1
q2
qx
P2
RAx
A
Qx
B
Me2
1 2 3 4 5 6 7 8 9 10
a a a a a a a a a a
RAy
RBy
10a
2a
RAy
P1
RAy-P1
q.a
2p
P2
P2
RBy
RBy-P2
MB
Me2
M1
M3
Menb.
M4
M34
M56
M6
M7
Me1
2p
3. p
RAx
Qx
qx.a

Reference:

All the figures are taken from the book given below:

- Prof. İbrahim EKİZ, YAPI STATİĞİ 1, BİRSEN YAYINEVİ